Sustainable Water Management

Spatial Data Analysis of the Water Quality in the Philadelphia Region

Shahryar Ali Zafar
Professor: Amanda Carneiro Marques
Vertically Integrated Project - ENGR-370

Abstract

This study focuses on spatial data analysis to investigate how different types of land use impact nearby water bodies in the Delaware River Basin. The `goal is to understand where and why water quality is being affected, using GIS tools to identify patterns and potential sources of degradation. Clean water is vital not only for human consumption but also for aquatic ecosystems and overall environmental health.

Our methodology included overlaying land use/land cover (LULC) data with watershed boundaries and categorizing land use into simple classification. We incorporated USGS weather station data, focusing on conductivity and dissolved oxygen (DO) levels, to assess water quality.

Key results showed that industrial zones had higher conductivity readings, likely due to runoff and pollutants, while residential areas near water bodies were associated with lower dissolved oxygen from possible nutrient loading or storm water discharge. These findings underline the importance of targeted land-use management and restoration efforts to improve water quality for drinking, aquatic life, and the broader ecosystem.

Table of Content

Section #	Section	Page Number
1	Introduction	3
2	Methods	3
3	Results	6
4	Discussion	8
5	Conclusion	9
6	References	10

1. Introduction

Water quality in the Philadelphia area and the greater Delaware River Basin has been a growing concern due to increasing urbanization, industrial activity, and runoff from residential zones. Pollutants like heavy metals, nutrients, and sediments enter local streams and rivers, leading to issues such as elevated conductivity, low dissolved oxygen (DO) levels, and harm to aquatic life. These degraded conditions not only impact drinking water sources but also affect the health of ecosystems and the communities that depend on them.

This study is important because water quality continues to decline and, if trends persist, is projected to worsen due to expanding land development and changing climate patterns. Water quality in the region has progressively declined and is projected to further deteriorate if proactive measures are not undertaken.

Our aim and objective is to analyze what factors are contributing to this decline, with a focus on different types of land use as a factor. The objective of this project is to use spatial data analysis to identify correlations between land use and water quality indicators and provide insights that support better land planning and environmental protection efforts.

2. Methods

To obtain the most accurate, up-to-date, and well-documented water quality data, we utilized datasets provided by the **United States Geological Survey** (USGS). These datasets originated from multiple USGS monitoring stations distributed across the Philadelphia region. The records span over a decade and include yearly, monthly, and weekly summaries. From these reports, we extracted key indicators of water quality—specifically, conductivity levels and dissolved oxygen (DO) percentages—and calculated average values from 11 monitoring stations to represent regional water conditions.

In parallel with water quality data, we sourced land use/land cover (LULC) and drainage basin data relevant to the Philadelphia area. To analyze the spatial distribution of land use types, we used data from **Open Data PHL**, which categorized land into residential, industrial, transportation, recreational, and other classifications. These categories were mapped based on both frequency and spatial area.

For watershed boundaries, our primary dataset came from **OpenDataPhilly**, which delineated the major drainage areas within Philadelphia. Figure 2.1 illustrates the overlay of land use distribution with these watershed boundaries. However, its limitation lies in the large spatial scale—each watershed covers a broad area, often encompassing multiple USGS stations, which made it more difficult to isolate localized land use impacts on water quality.

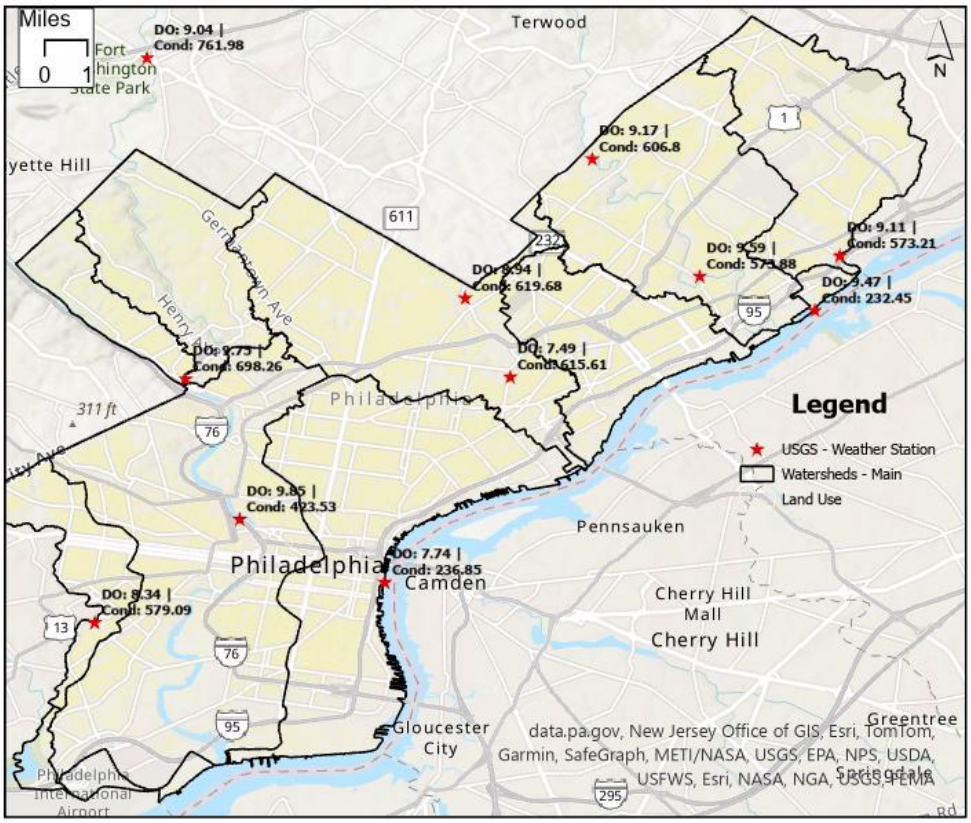

Figure 2.1

To improve spatial resolution, we integrated a more detailed watershed dataset from **PASDA** (Pennsylvania Spatial Data Access). This dataset provided refined watershed boundaries across Pennsylvania. We clipped it to match the extent of our land use data, creating smaller, more localized drainage sub-basins for targeted analysis. Figure 2.2 displays land use categories within these finer watershed divisions, allowing for more precise correlation between land use and water quality metrics.

While the PASDA dataset enabled a more focused spatial analysis, the broader Open Data PHL watershed dataset offered greater coverage, especially in areas that included key USGS stations. To ensure consistency and reliability, we conducted land use analysis across both datasets, enabling cross-verification of results.

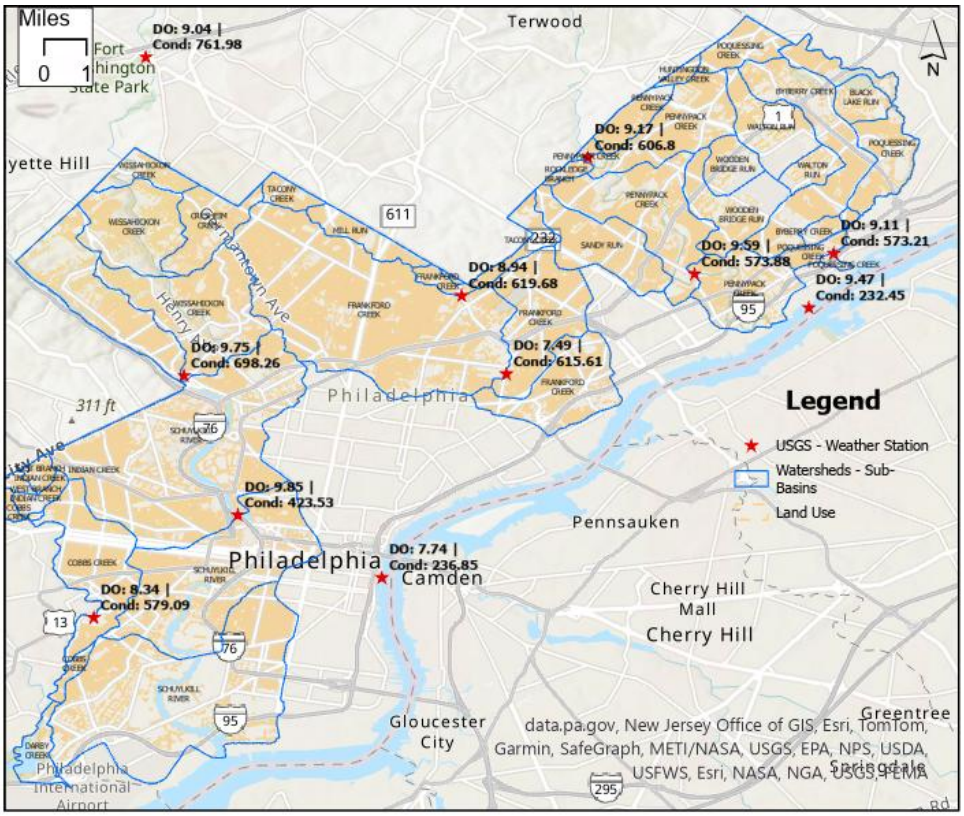

Figure 2.2

All datasets were imported into ArcGIS Pro, where we performed spatial operations—including clipping and spatial joins—to calculate the percentage of each land use type within each watershed. This integrated GIS approach allowed us to effectively examine the relationship between land use distribution and changes in water conductivity and dissolved oxygen across the Philadelphia region.

3. Results

Our methods to collect and process spatial and water quality data led us to a clear picture of how land use types relate to water conditions in Philadelphia's watersheds. By combining GIS-based land use mapping with station-level conductivity and dissolved oxygen readings, we were able to quantify and visualize these relationships. The following results summarize these findings through both data tables and graphical representations.

Table 3.1 highlights key watersheds that have USGS weather stations located within or near their boundaries. In *Figure 2.1* and *Figure 2.2*, some stations fall outside the Philadelphia region, and thus were excluded from the table. Additional exclusions were made for stations that appeared multiple times within the same watershed—only the downstream station was retained to better represent cumulative water quality conditions. The table also presents the corresponding percentages of residential and industrial land use within each watershed.

Table 3.1

Station	Watersheds	Dissolved O_2 Level (mg/L)	Conductivity Level (µS/cm)	Residential Land %	Industrial Land %
1467042	PENNY PACK CREEK	9.17	606.81	39.63	6.13
1467086	TACONY CREEK	8.94	619.68	14.47	0.15
1474000	WISSAHICON CREEK	9.75	698.26	37.95	0.21
1475548	COBBS CREEK	8.34	579.09	48.58	2.47
1474500	SCHUYLKILL RIVER	9.85	423.53	20.62	11.85
1467087	FRANKFORD CREEK	7.49	615.61	48.76	11.05
1465798	POQUESSING CREEK	9.11	573.21	23.93	2.35
1467200	DELAWARE DISTRICT	7.74	236.85	75.81	1.43

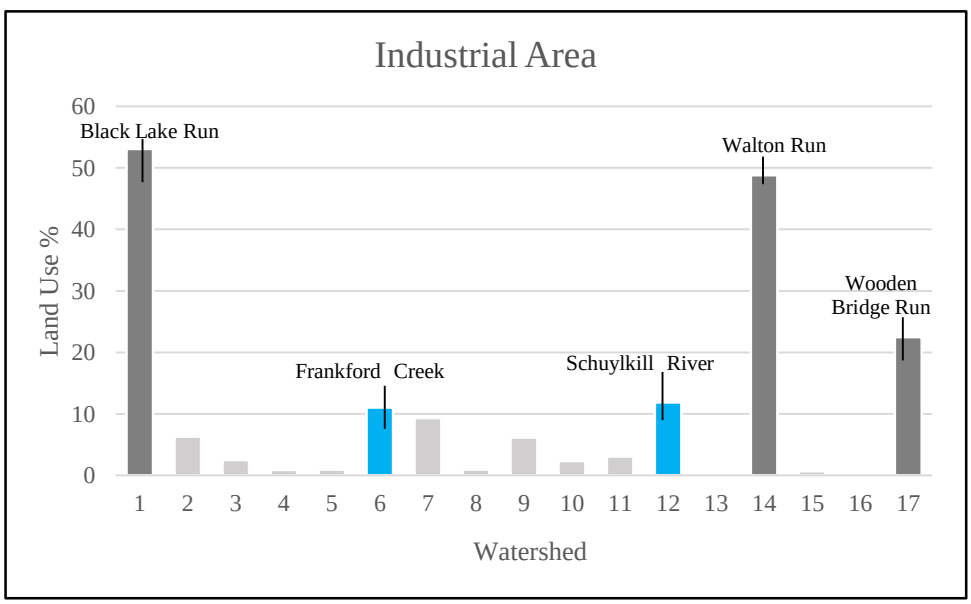

Figure 3.2

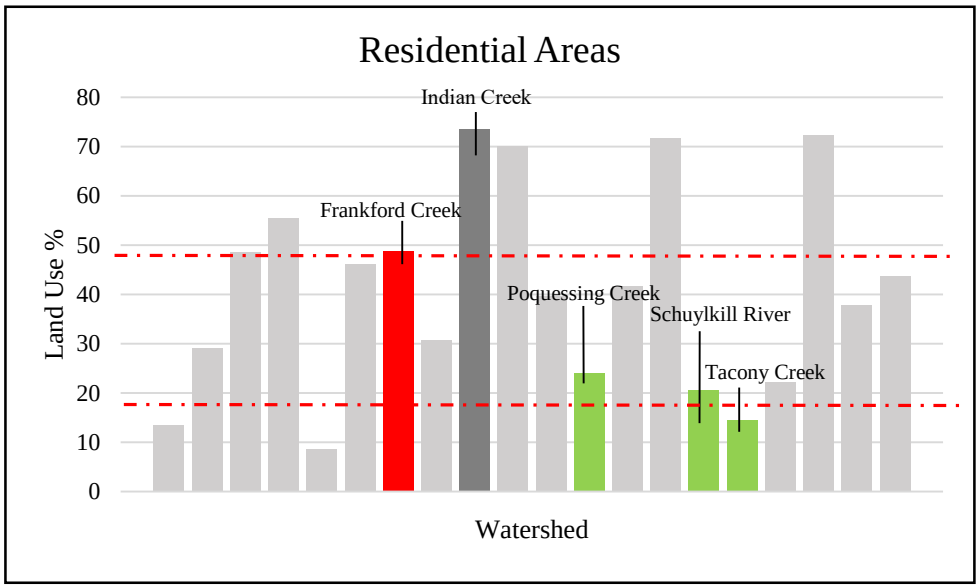

Figure 3.3

Figure 3.2 and *Figure 3.3* provide a simplified visual breakdown of the data presented in the table. They illustrate various watersheds and compare their respective land use and land cover distributions. Select watersheds are labeled and highlighted to emphasize key findings that align with the analysis and support the discussion.

Figure 3.2 illustrates the watersheds and the proportion of industrial land use within each. **Black Lake Run**, **Walton Run**, and **Wooden Bridge Run** display the highest percentages of industrial area; however, they are excluded from further analysis due to the absence of USGS stations within their drainage areas. Following these, the **Frankford Creek** and **Schuylkill River** watersheds exhibit the next highest industrial land use, with approximately **11%** of their areas classified as industrial. Frankford Creek reports a conductivity value of **615.61 µS/cm**, while Schuylkill River reports the lowest among the group, at **423.53 µS/cm**.

Figure 3.3 presents the same data format as *Figure 3.2*, but focuses on residential land use across the watersheds. Some watersheds, such as **Indian Creek**, exhibit high residential coverage but are excluded from the analysis due to the absence of reporting weather stations. Among the remaining highlighted watersheds, the data is categorized into two groups. The first includes **Frankford Creek**, with **48.76%** of its area designated for residential use. The second group consists of **Poquessing Creek**, **Schuylkill River**, and **Tacony Creek**, each with less than **25%** residential land use. Frankford Creek reports the lowest dissolved oxygen level at **7.49 mg/L**, while the other three watersheds show DO levels ranging from **8.94 mg/L** to **9.85 mg/L**.

4. Discussion

This study aimed to analyze how different types of land use—specifically residential and industrial—impact water quality, measured through dissolved oxygen (DO) and conductivity levels. Based on our hypothesis, higher industrial land use would correlate with elevated conductivity, due to increased runoff containing salts, chemicals, and pollutants. Similarly, higher residential land use was expected to correlate with lower dissolved oxygen, likely due to organic matter, fertilizers, and nutrient loading from urban runoff.

To contextualize our findings, it's important to note that optimal dissolved oxygen levels for healthy aquatic ecosystems are generally above **8.0 mg/L**, and conductivity values between **150–500 µS/cm** are considered ideal for freshwater streams *(EPA, 2022; USGS, 2020)*. In our study, several watersheds exceed or fall below these thresholds. For example, Frankford Creek showed a high residential land use (**48.76%**) and recorded the lowest DO level (**7.49 mg/L**) among the evaluated watersheds. This supports our hypothesis that residential development contributes to oxygen depletion. In addition to runoff, urban sewer overflows during heavy rainfall can lead to discharge untreated or partially treated waste into water bodies, further reducing dissolved oxygen levels by increasing biological oxygen demand (BOD).

Conversely, Schuylkill River, with 11.85% industrial land use, exhibited the lowest conductivity value (**423.53 µS/cm**), contrary to expectations. This does not align with a general following trend. However, it's worth noting that this watershed spans a larger and more hydrologically diverse area, which may dilute pollutant concentrations. Frankford Creek, also high in industrial land use, recorded a conductivity level of **615.61 µS/cm**, which supports our assumption regarding industrial impacts.

A 2020 assessment by the USGS linked elevated conductivity in urban streams to roadway salts and industrial discharge. Our findings, particularly in **Poquessing Creek** and **Tacony Creek**, reinforce this connection: these watersheds, with moderate land use intensity, showed moderate conductivity and relatively healthy DO levels, suggesting that less disturbed areas maintain better water quality balance.

It is important to acknowledge a key limitation of this study. It is that land use patterns have changed over time, yet some of the watershed delineations and datasets represent current or aggregated land use. This temporal mismatch means water quality readings from previous years may not reflect the exact land cover conditions at the time. Despite this, the patterns observed still demonstrate clear associations between land use type and water quality trends, especially when averaged over long-term datasets.

5. Conclusion

This study highlights the strong connection between land use and water quality in the Philadelphia region. Using USGS station data and spatial land use mapping, we analyzed how residential and industrial development influence dissolved oxygen and conductivity levels across local watersheds.

Our results support the hypothesis that residential areas tend to have lower dissolved oxygen, likely due to nutrient runoff and sewer overflows, while industrial zones generally exhibit higher conductivity, indicating increased pollutant levels. Watersheds with higher percentages of these land uses showed more significant impacts on water quality, reinforcing the relationship between land activity and ecological conditions.

Ultimately, understanding how land use affects water quality is essential for protecting aquatic ecosystems, ensuring safe drinking water, and promoting the overall health of communities and the environment. These insights can help guide smarter development and long-term sustainability for both people and nature.

6. References

Studies Quoted;

EPA (2022). *Water Quality Standards Handbook: Chapter 3 — Dissolved Oxygen, Temperature, and pH.* United States Environmental Protection Agency. https://www.epa.gov/wqs-tech

USGS (2020). *Water Quality Characteristics of U.S. Streams: Conductivity and Dissolved Solids.* United States Geological Survey. Retrieved from https://waterdata.usgs.gov

Datasets for Shape Files;

https://data-phl.opendata.arcgis.com/

https://opendataphilly.org/datasets/major-watersheds-philadelphia/

https://www.pasda.psu.edu/uci/DataSummary.aspx?dataset=14

https://www.usgs.gov/data/philadelphia-area-groundwater-level-network